

Solution of Tutorial-3

Engineering Mechanics: ME-101 (Semester-II, 2024)

Time: 7:55 – 8:50 AM

Q1. Determine the moment M which must be applied at A to keep the frame in static equilibrium in the position shown in **Figure 1**. Also calculate the magnitude of the pin reaction at A . Assume the pulley and links are massless, and all the joints are frictionless.

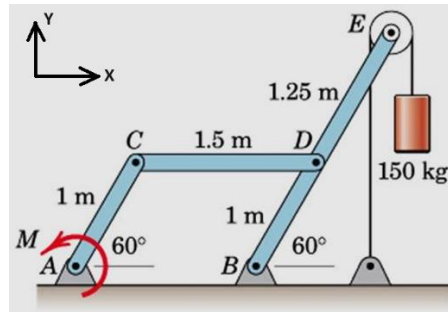
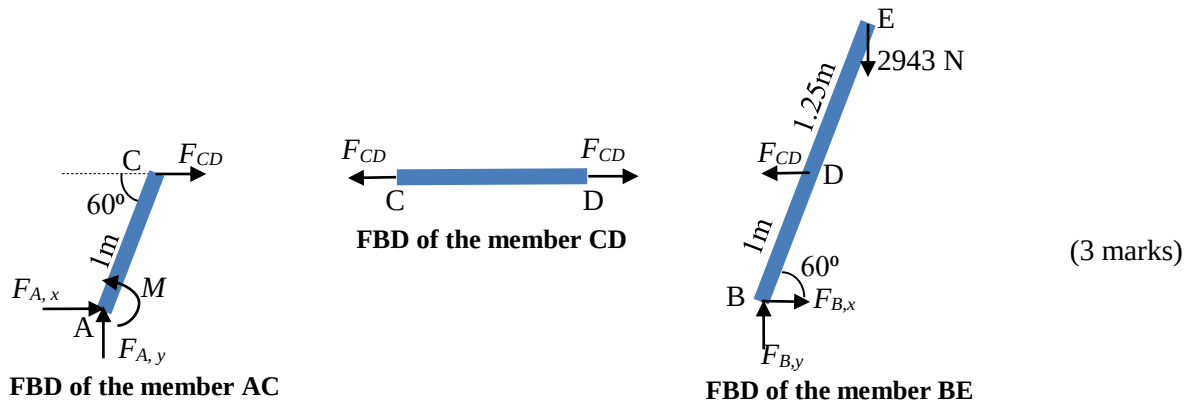


Figure 1

Solution of Q1:

The pulley is carrying a weight of $150 \times g$, i.e. 1471.5 N.

Due to the weight and the tension in the rope, there will be a vertical reaction at point E as shown in figure. (1 mark)



Since, ABDC is a parallelogram, the member CD will remain horizontal at all times. Therefore, there will only be horizontal reaction from CD on the member BE.

Considering the FBD of the member BE, as shown above, the moment equilibrium equation about point B:

$$\sum M_B = 0$$

$$\Rightarrow (F_{CD} \times 1 \sin 60^\circ) - (2943 \times 2.25 \cos 60^\circ) = 0$$

$$\Rightarrow F_{CD} = 3823.07 \text{ N} = 3.82 \text{ kN (tensile in member CD)} \quad (2 \text{ mark})$$

Now, consider the FBD of member AC. Considering the moment equilibrium about pin A:

$$\sum M_A = 0$$

$$\Rightarrow M - F_{CD} \times 1 \sin 60^\circ = 0$$

$$\Rightarrow M = 3.31 \text{ kN-m}$$

(2 marks)

Invoking force equilibrium equations for member AC:

$$\sum F_y = 0$$

$$\Rightarrow F_{A,y} = 0 \text{ kN}$$

$$\text{and } \sum F_x = 0$$

$$\Rightarrow F_{CD} + F_{A,x} = 0$$

$$\Rightarrow F_{A,x} = -3.82 \text{ kN}$$

$$\text{Therefore, reaction at pin A} = \sqrt{F_{A,x}^2 + F_{A,y}^2} = F_{A,x} = 3.82 \text{ kN}$$

(2 marks)

Q2. Replace the distributed loading on the beam shown in **Figure 2** with an equivalent resultant force, and specify its location on the beam measured from point A.

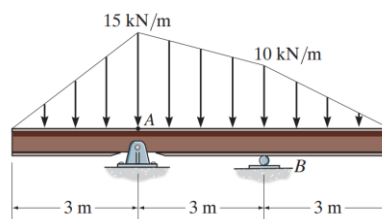
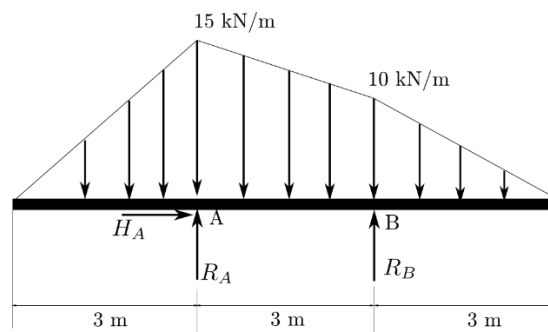


Figure 2

Solution of Q2: The free body diagram of the beam is illustrated below.



(1 mark)

Step 1: Finding out the support reactions

Since, there is no horizontal loading, $\sum F_x = 0$ leads to $H_A = 0$

Now, $\sum F_y = 0$, (considering upward forces positive)

$$\Rightarrow R_A + R_B - \left[\left(\frac{1}{2} \times 3 \times 15 \right) + \left\{ \frac{1}{2} \times (15 + 10) \times 3 \right\} + \left(\frac{1}{2} \times 3 \times 10 \right) \right] = 0$$

$$\Rightarrow R_A + R_B = 75 \quad \text{----- (1)}$$

(2 marks)

$\sum M_A = 0$, (considering counter clockwise moments positive)

$$\Rightarrow R_B \times 3 + \left\{ \left(\frac{1}{2} \times 3 \times 15 \right) \times 1 \right\} - \left\{ \left(\frac{1}{2} \times 3 \times 10 \right) \times 4 \right\} - \left\{ \left(\frac{1}{2} \times (15 + 10) \times 3 \right) \times \frac{7}{5} \right\} = 0$$

$$\Rightarrow R_B = 30 \text{ kN}$$

From equation (1), we obtain $R_A = 45 \text{ kN}$

(2 marks)

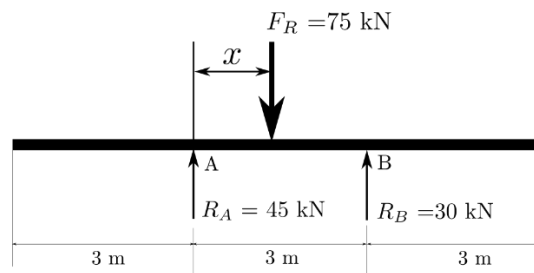
Step 2: Finding out the equivalent resultant force and its location

The equivalent resultant force,

$F_R = \text{Total area of the distributed loading diagram}$

$$= [(1/2 \times 3 \times 15) + \{1/2 \times (15 + 10) \times 3\} + (1/2 \times 3 \times 10)] = 75 \text{ kN (downward)} \quad (2.5 \text{ marks})$$

Let the line of action of the equivalent resultant force (F_R) be x from support A as shown below



Now, $\sum M_A = 0$, (considering counter clockwise moments positive)

$$\Rightarrow R_B \times 3 - F_R \times x = 0 \quad \Rightarrow 30 \times 3 - 75 \times x = 0 \quad \Rightarrow x = 1.2 \text{ m (in the right of A)}$$

(2.5 marks)

Q3. For the shaded area as shown in **Figure 3**, calculate the following:

- area of the shaded portion
- x - coordinate of centroid of shaded area in terms of a and b
- y - coordinate of centroid of shaded area in terms of a and b
- the co-ordinates of the centroid when $a = 50 \text{ mm}$ and $b = 40 \text{ mm}$

[3 marks]

[3 marks]

[3 marks]

[1 mark]

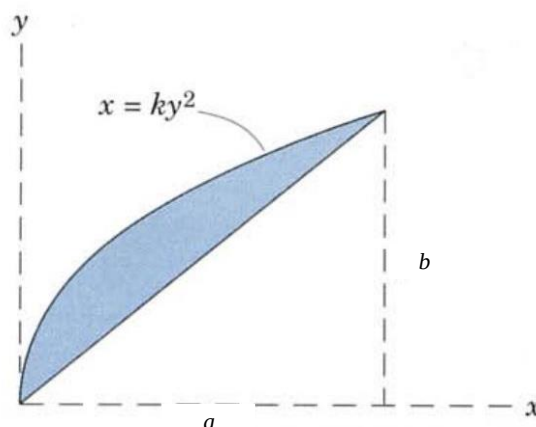
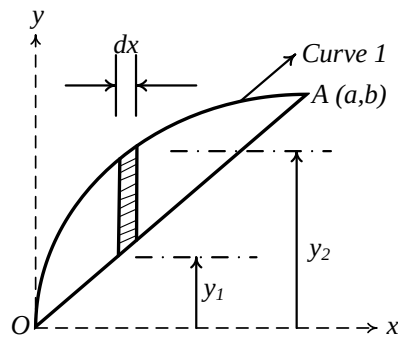


Figure 3

Solution of Q3:**Step 1: Calculation of area of shaded portion****[3 marks]****Figure 3.1**

Equation of *curve 1* is $x = ky^2$

For *curve 1* value of k : $a = kb^2 \rightarrow k = \frac{a}{b^2}$

Let the equation of straight line *OA* is $x = k_1y$

For straight line *OA* value of k_1 : $a = k_1b \rightarrow k_1 = \frac{a}{b}$

From *curve 1* value of y_2 :

$$y_2 = \sqrt{\frac{x}{k}} = b\sqrt{\frac{x}{a}}$$

From straight line *OA* value of y_1 :

$$y_1 = \frac{x}{k_1} = \frac{b}{a}x$$

Area of the shaded strip, $dA = (y_2 - y_1)dx$

Total area of the shaded portion

$$\begin{aligned} A &= \int_0^a (y_2 - y_1)dx \\ &= \int_0^a \left(b\sqrt{\frac{x}{a}} - x\frac{b}{a} \right) dx \\ &= b \left[\frac{1}{\sqrt{a}} \frac{2x^{\frac{3}{2}}}{3} - \frac{1}{2a} x^2 \right]_0^a \\ &= \frac{ab}{6} \end{aligned}$$

Step 2: Calculation of x- coordinate of centroid of shaded area in terms of a and b
[3 marks]

Taking first moment of area about y- axis

$$\begin{aligned}\int x_c dA &= \int_0^a x(y_2 - y_1) dx \\ &= \int_0^a \left[\frac{b}{\sqrt{a}} x^{\frac{3}{2}} - \frac{b}{a} x^2 \right] dx \\ &= b \left[\frac{2x^{\frac{5}{2}}}{5\sqrt{a}} - \frac{x^3}{3a} \right]_0^a \\ &= \frac{a^2 b}{15}\end{aligned}$$

x co-ordinate of CG of the shaded area

$$\begin{aligned}\bar{x} &= \frac{\int x_c dA}{A} \\ \bar{x} &= \frac{\frac{a^2 b}{15}}{\frac{ab}{6}} \\ \bar{x} &= \frac{2}{5} a\end{aligned}$$

Step 3: Calculation of y- coordinate of centroid of shaded area in terms of a and b
[3 marks]

Taking first moment of area about x- axis

$$\begin{aligned}y_c dA &= \int_0^a \left(\frac{y_1 + y_2}{2} \right) (y_2 - y_1) dx \\ &= \frac{1}{2} \int_0^a (y_2^2 - y_1^2) dx \\ &= \frac{1}{2} \int_0^a \left(\frac{xb^2}{a} - \frac{x^2 b^2}{a^2} \right) dx \\ &= \frac{1}{12} ab^2\end{aligned}$$

y co-ordinate of CG of the shaded area

$$\bar{y} = \frac{y_c dA}{A}$$

$$\bar{y} = \frac{ab^2}{\frac{ab}{6}}$$

$$\bar{y} = \frac{b}{2}$$

Step 4: When $a = 50 \text{ mm}$ and $b = 40 \text{ mm}$, then co-ordinate of the centroid [1 mark]

$$\bar{x} = \frac{2}{5}a = 20 \text{ mm}$$

$$\bar{y} = \frac{b}{2} = 20 \text{ mm}$$

Q4. Figure 4 shows a concrete dam, whose factor of safety for tipping is defined as the ratio of the stabilizing moment due to the dam's weight divided by the overturning moment about O due to the water pressure.

- 1) Draw the Free Body Diagram of the dam. [2 Mark]
- 2) Calculate the pressure at the base of the dam and intensities of the load at the base of the dam [1 Mark]
- 3) Calculate the dam's weight and the point of application [3 Marks]
- 4) Calculate the force due to water pressure and the point of application [2 Marks]
- 5) Calculate the overturning moment and stabilizing moment [2 Marks]

The concrete has a density $\rho_c = 2.5 \text{ Mg/m}^3$ and for water the density is $\rho_w = 1 \text{ Mg/m}^3$.

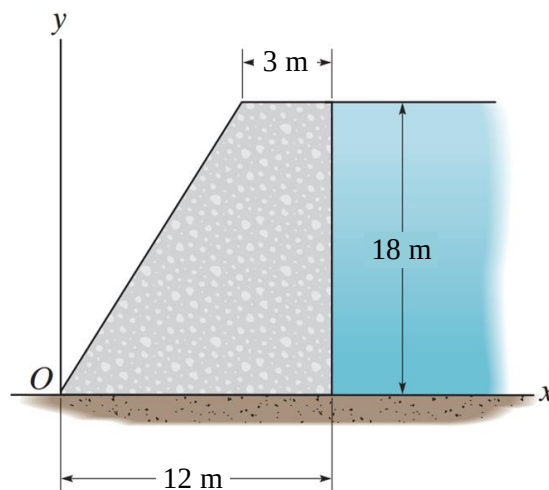


Figure 4

Solution of Q4:

Step 1: Drawing the Free Body Diagram of given figure.

[2 Mark]

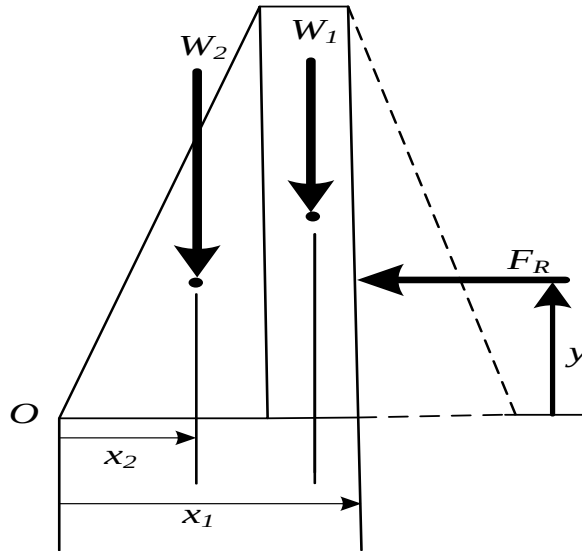


Figure 4.1

(NOTE: Someone may represent the total weight of the dam and illustrate it as one force passing through the centroid of the trapezoid. In that case, either he has to derive or mention explicitly the coordinate of the centroid.)

Step 2: Calculation of the pressure at the base of the dam and intensities of the load at the base of the dam [2 Marks]

The computation will be based on $b = 1$ m width of the dam. The pressure at the base of the dam is

$$P = \rho_w g h = 1000(9.81)(18) = 176.58(10^3) \text{ Pa} = 176.58 \text{ kPa}$$

Thus, since the dam has a constant width, the pressure loading can be viewed in two dimensions as shown in Figure 4.1. The intensities of the load at the base of the dam is-

$$w = Pb = 176.58(1) = 176.58 \text{ kN/m}$$

Step 3: Calculation of the dam's weight and the point of application [2 Marks]

$$\text{Weight} = \text{density} \times \text{volume} \times \text{acceleration due to gravity}$$

The weight of the dam and the distance of the respective points of application from O, shown in Figure 4.1, are

$$W_1 = (2500)[(3)(18)(1)](9.81) = 1324.35(10^3) \text{ N} = 1324.35 \text{ kN}$$

$$x_1 = 9 + \frac{1}{2}(3) = 10.5 \text{ m}$$

$$W_2 = (2500) \left[\frac{1}{2}(9)(18)(1) \right] (9.81) = 1986.525(10^3) \text{ N} = 1986.525 \text{ kN}$$

$$x_2 = \frac{2}{3}(9) = 6 \text{ m}$$

The total weight of the dam is 3310.88 kN, and is acting through a point located at 7.8 m from O .

Step 4: Calculation of the force due to water pressure i.e. F_R and the point of application [2 Marks]

The force due to water pressure is given by,

$$F_R = \frac{1}{2}(176.58)(18) = 1589.22 \text{ kN}$$

Also, the distance of the points of application from base is given by

$$y = \frac{1}{3}(18) = 6 \text{ m (from base)}$$

Step 5: Calculation of the overturning moment and stabilizing moment [2 Marks]

The stabilizing moment is due to the dam's weight i.e. $(W_1 + W_2)$ and the overturning moment is due to the water pressure i.e. F_R .

Thus, the overturning moment about O is

$$M_{OT} = 1589.22(6) = 9535.32 \text{ kN} \cdot \text{m}$$

And the stabilizing moment about O is

$$M_s = 1324.35(10.5) + 1986.525(6) = 25824.825 \text{ kN} \cdot \text{m}$$